

TEA018 - Hidrologia Ambiental

Dados de evapotranspiração de satélite - MODIS/NASA

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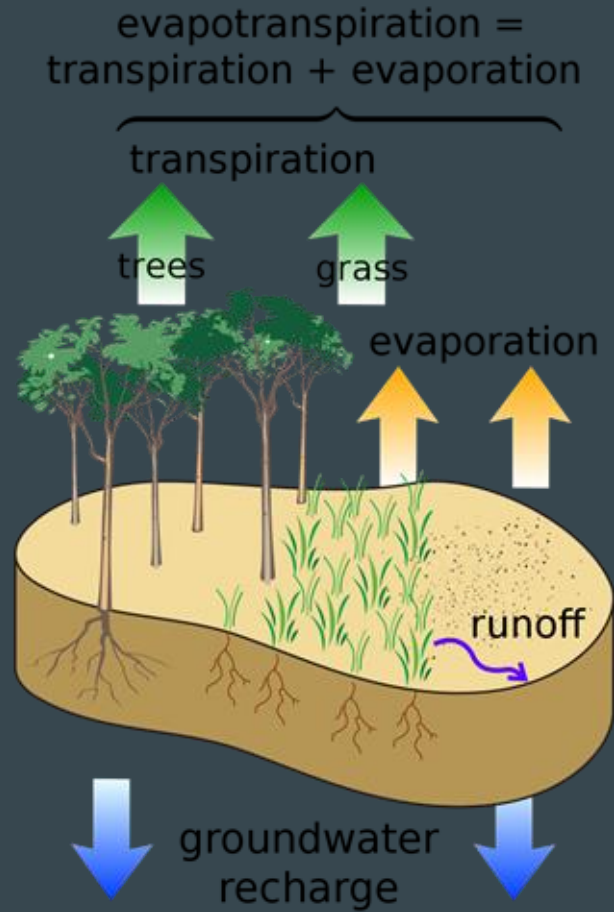
Agenda

- O que é Evapotranspiração?
- Balanço hídrico em bacias hidrográficas, Fotossíntese florestal
- Aplicativo de extração e exploração de amostras para análise (AppEEARS)
- Baixe os dados
- Importe os dados para o Google Colab
- Reamostrar os dados
- Trace e compare diferentes bacias hidrográficas

O que é Evapotranspiração?

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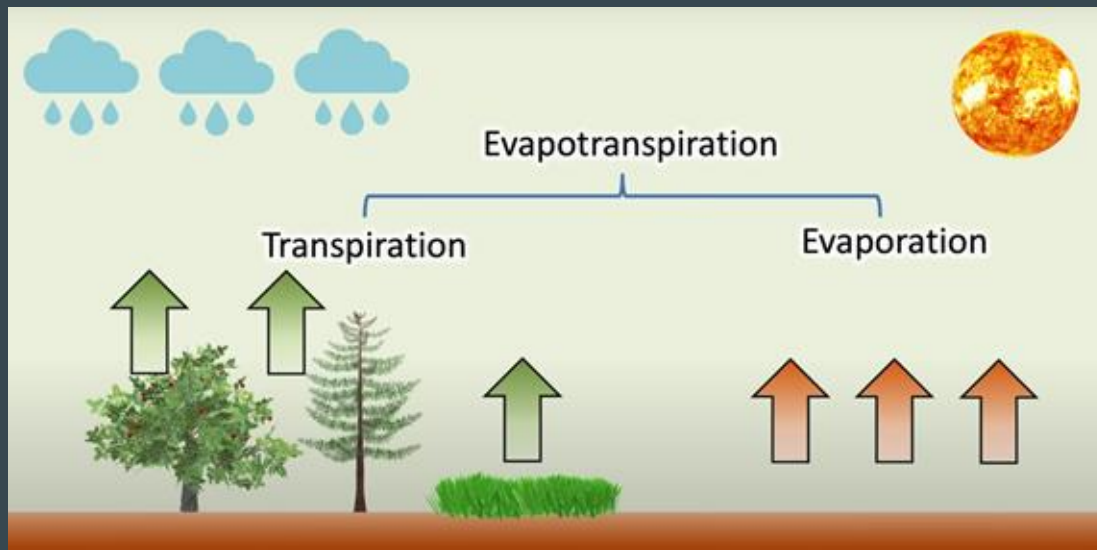
- Transpiração + Evaporação
- Evapotranspiração potencial x real



O que é evapotranspiração?

Forçantes da evapotranspiração:

- Radiação solar
- Disponibilidade de água no solo/planta
- Gradiente de vapor de água no ar
- Transporte de vapor de água pelo vento



Introduction to MODIS Evapotranspiration (MOD16) - a free global dataset of ET & PET
https://www.youtube.com/watch?v=3r_6il0EViw

Nas plantas vasculares, a água sai da planta através dos estômatos nas folhas, enquanto que nas plantas não vasculares (briófitas, musgos e algas), ela sai através dos filídeos.

https://en.wikipedia.org/wiki/Non-vascular_plant

Transpiração em plantas vasculares

- plants retain less than 5% of water absorbed by roots for growth.

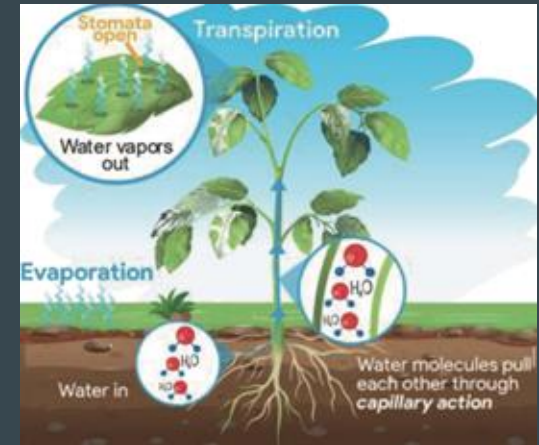


it goes back to the atmosphere!

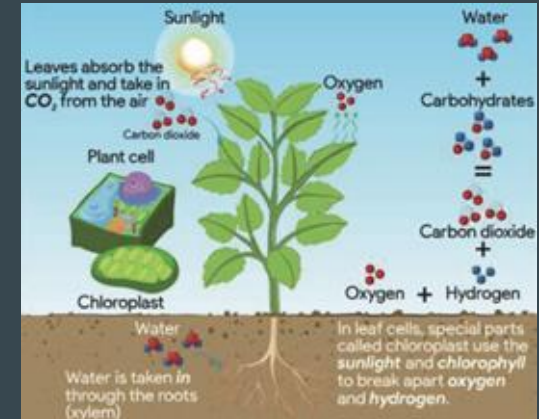
Photosynthesis



- To make sugars, plants must absorb carbon dioxide (CO₂) from the atmosphere through **stomata**.

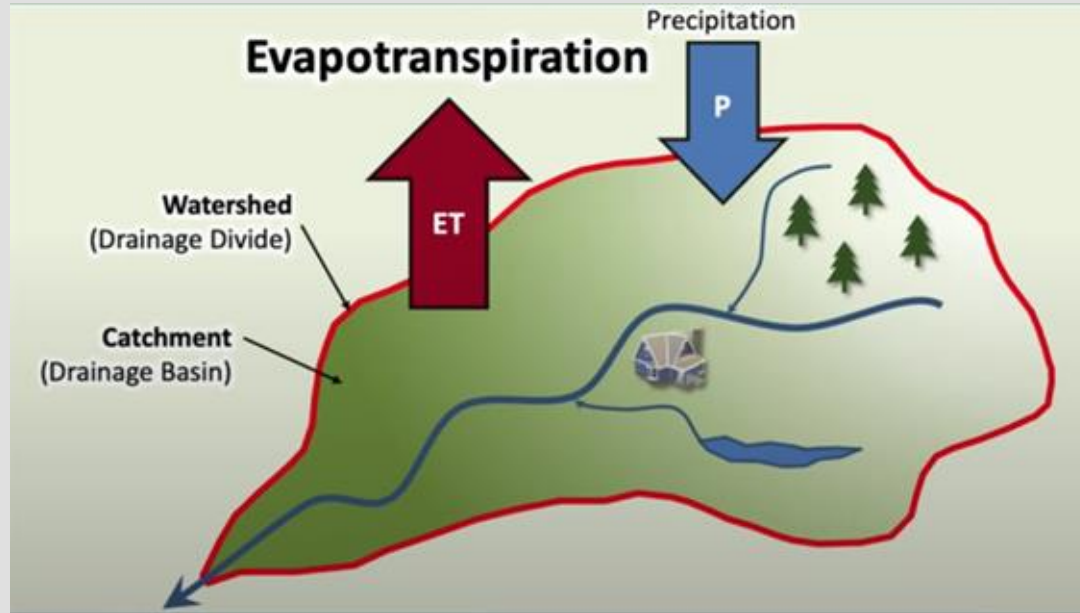


transpiration



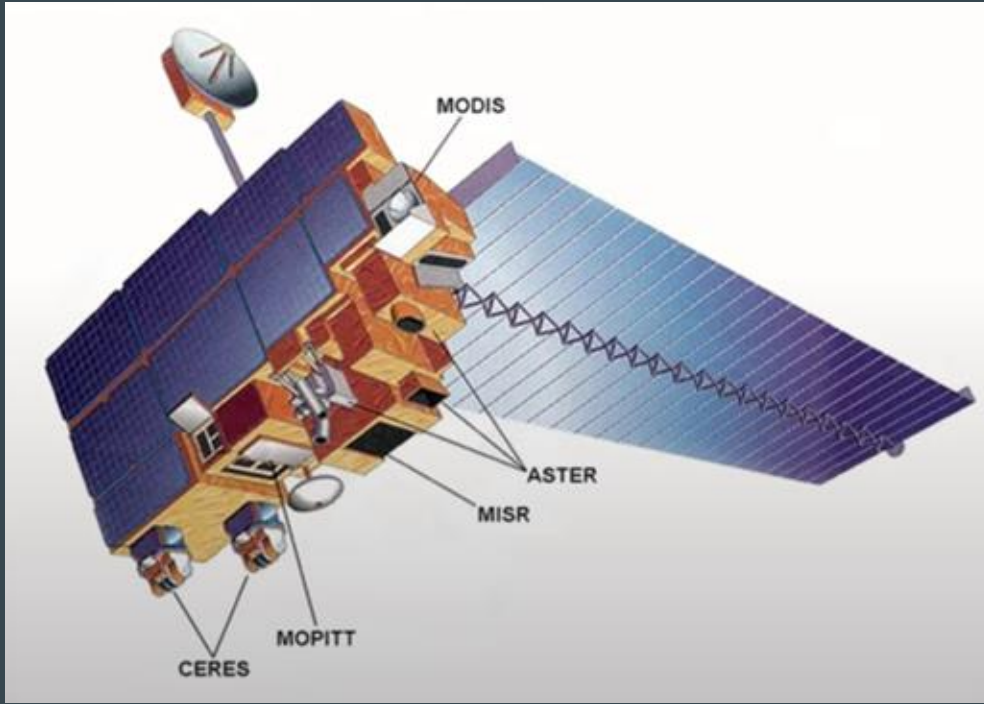
photosynthesis

Water balance in river basins



Introduction to MODIS Evapotranspiration (MOD16) - a free global dataset of ET & PET
https://www.youtube.com/watch?v=3r_6il0EViw

What is MODIS?



- Instrument on board of TERRA & AQUA satellites (NASA)
- TERRA = "MOD"
- AQUA = "MYD"

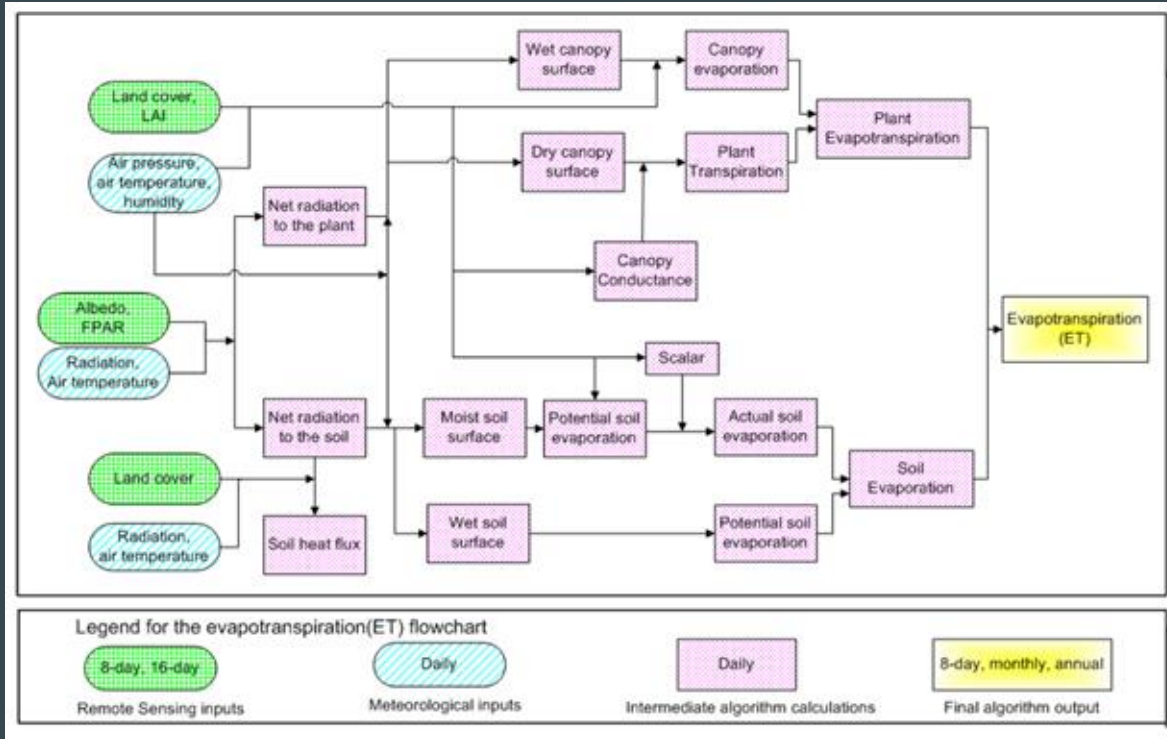
MODIS MOD16 ET Product

- ❖ Evapotranspiration (ET) →
 - Total ET
 - Total PET
- ❖ Latent Heat Flux (LE)
- ❖ Quality control flags

MODIS MOD16 ET Product

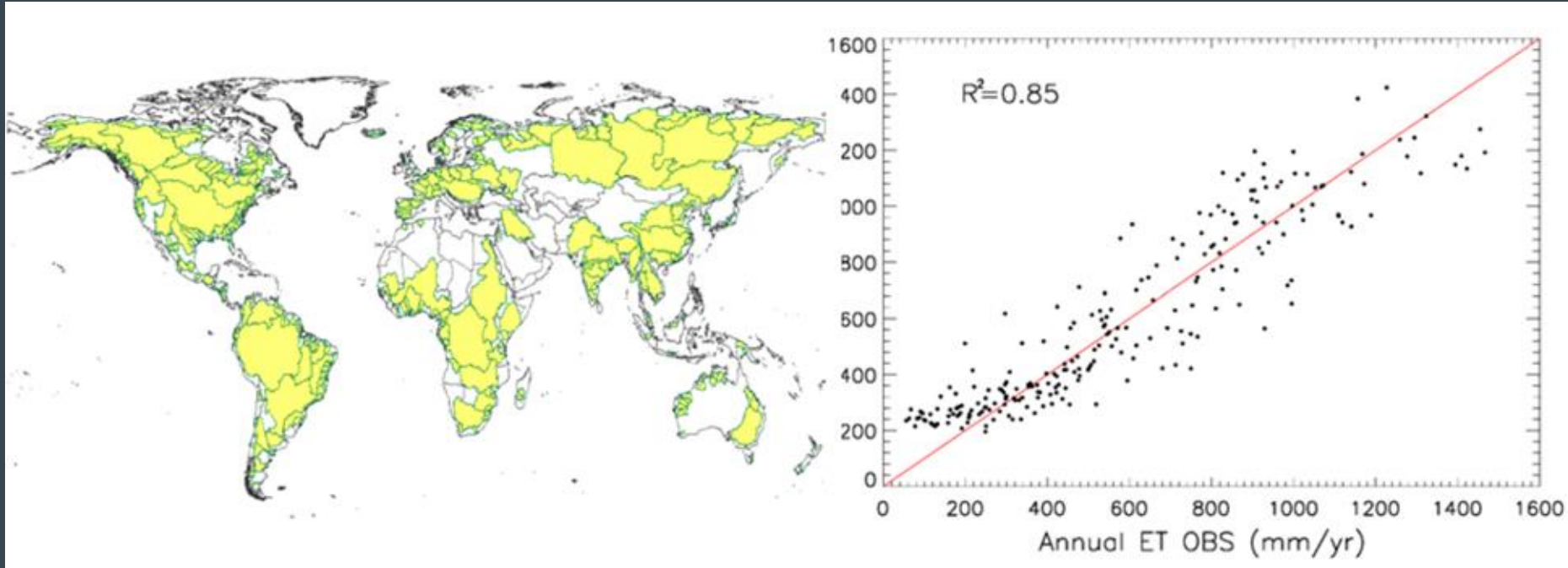
- O algoritmo MODIS ET segue a equação de Penman-Monteith.
- Inclui a evaporação do solo úmido e molhado, a evaporação da água da chuva interceptada pela copa das plantas antes de atingir o solo e a transpiração através dos estômatos nas folhas e caules das plantas.
- Os produtos MOD16A2/A3 ET são produzidos em intervalos de 8 dias e anuais.

Flowchart of the improved MOD16 ET algorithm.



LAI: leaf area index; FPAR: Fraction of Photosynthetically Active Radiation.

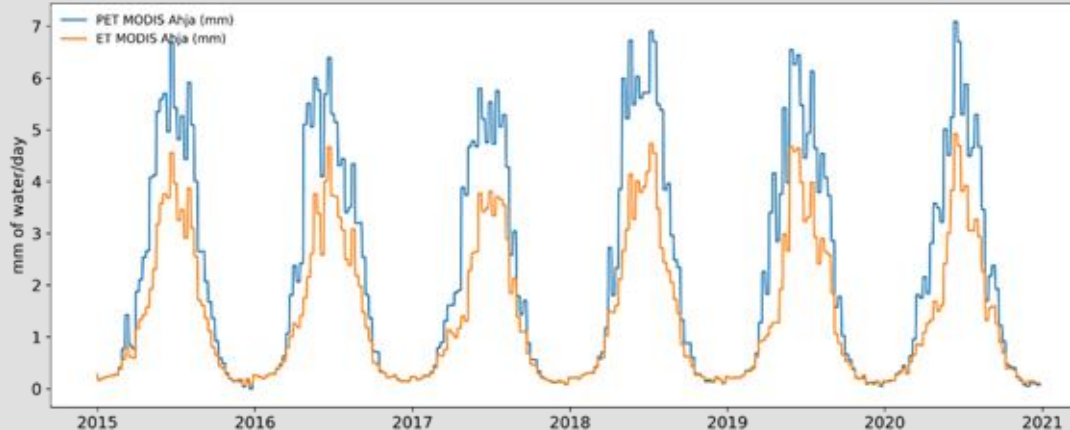
Algorithm Performance at Global Watersheds



The MOD16 ET estimates can explain 85% of the variations of the pseudo-ET observations for 232 river basins.

Time series: 2000 ...

- Global evapotranspiration data set
- Spatial resolution: 500 x 500 m
- View some ET and PET data from Ahja river basin, Estonia



Search Data Catalog



The Land Processes Distributed Active Archive Center (LP DAAC) is one of several discipline-specific data centers within the NASA Earth Observing System Data

<https://lpdaac.usgs.gov/>

Search Data Catalog

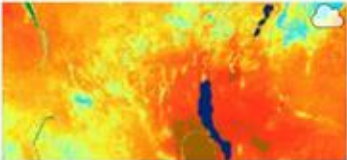
What are you looking for...

Search

Cloud Access Temporal Range Collection Version Keyword Spatial Resolution

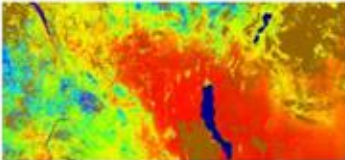
KEYWORD: EVAPOTRANSPIRATION (ET) x SPATIAL RESOLUTION (M): 500.0 x STATUS: OPERATIONAL x [Clear filters](#)

Showing 1 - 14 of 14 results



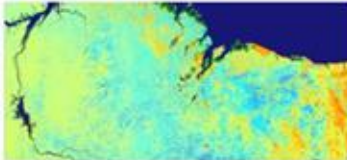
MOD16A2 v061
MODIS/Terra Net Evapotranspiration 8-Day L4 Global 500 m SIN Grid

TERRA MODIS EVAPOTRANSPIRATION (ET)



MOD16A2 v006
MODIS/Terra Net Evapotranspiration 8-Day L4 Global 500 m SIN Grid

TERRA MODIS EVAPOTRANSPIRATION (ET)



MOD16A2GF v061
MODIS/Terra Net Evapotranspiration Gap-Filled 8-Day L4 Global 500 m SIN Grid

TERRA MODIS EVAPOTRANSPIRATION (ET)

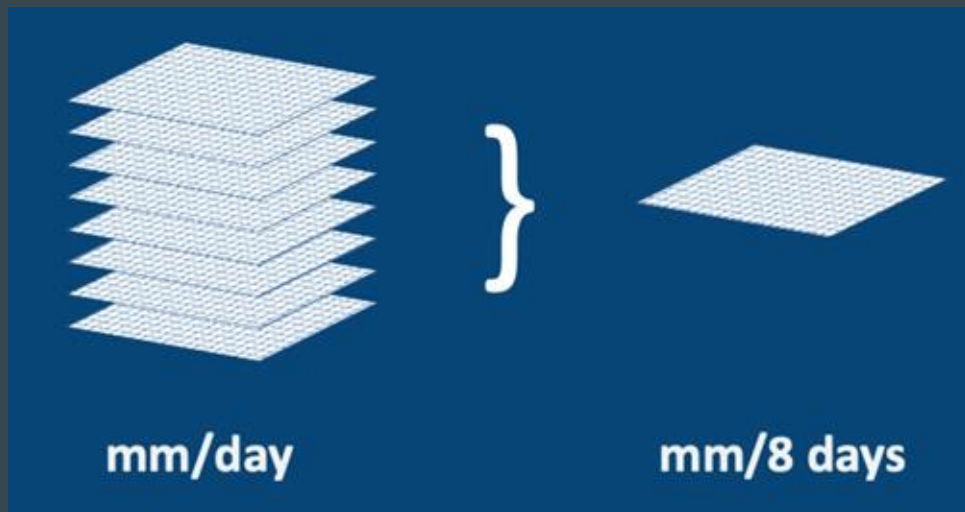
MOD16A2.* and MOD16A2GF.* files

<https://lpdaac.usgs.gov/>

MODIS MOD16A2GF Version 061

- Global ET & PET dataset
- Spatial resolution: 500 m x 500 m
- Time series: January 2000 - present
- Time steps: 8-day composite

8-day composite



Source: Introduction to MODIS Evapotranspiration (MOD16) - a free global dataset of ET & PET
https://www.youtube.com/watch?v=3r_6il0EViw

AppEEARS - sign in!



Application for Extracting and Exploring Analysis Ready Samples (AppEEARS)

<https://appeears.earthdatacloud.nasa.gov/>

AppEEARS - Downloading the ET data

- Access the GitHub of the class:

<https://github.com/emiliomercuri/hidrologia/>

Download 3 Shapefiles (perimeter of the water basins)

- Nhundiaquara river (Paraná - Brazil)
 - Ahja river (Estonia)
 - Salma river (Afghanistan)
- Download the 3 zip files!
 - Use the shapefile from your watershed

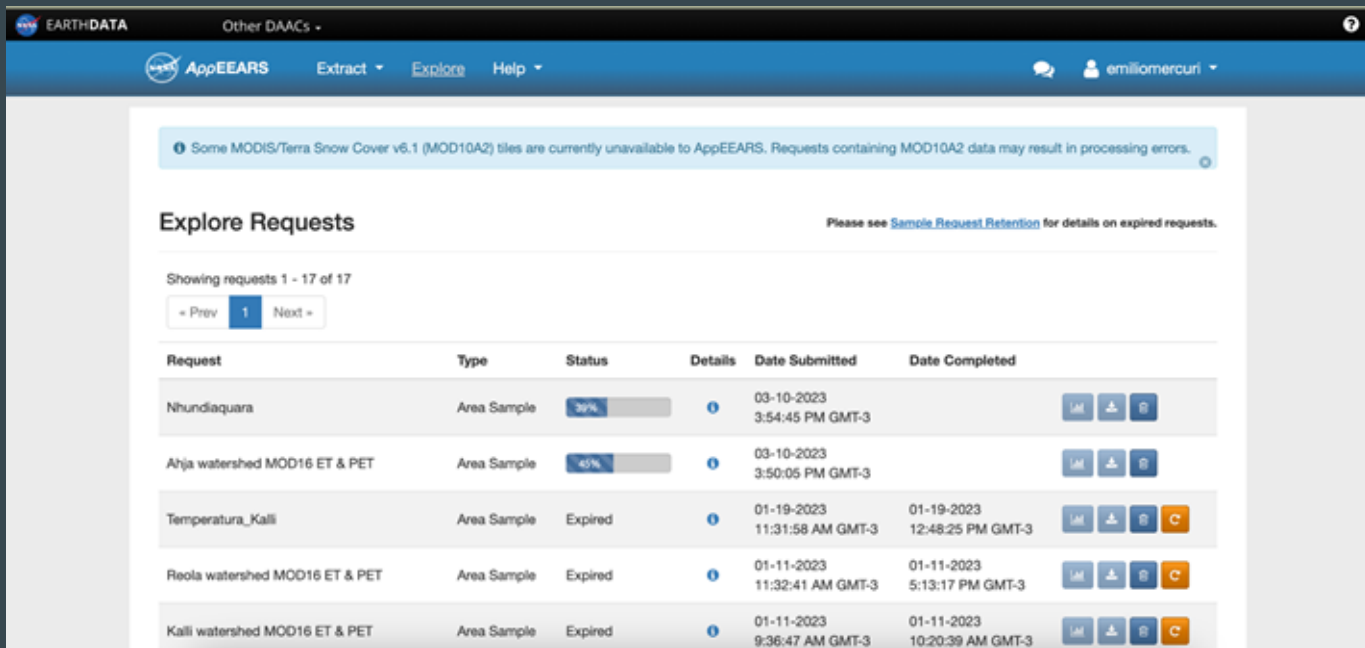
Access AppEEARS - Downloading the ET data

- Extract -> Area -> Start a new request
 - Enter a name to identify your sample: Purus
 - Drop a vector polygon: shapefile.zip (Gerado na aula 2 - Colab 2024_03_01_Atividade_em_aula_delineamento.ipynb)
 - Select the layer: MOD16A2GF.061

The screenshot shows the AppEEARS web application interface. The top navigation bar includes 'Extract', 'Explore', and 'Help' tabs. The main content area is divided into several sections:

- Upload a file or draw a polygon using the [icon] or [icon] icon:** A dashed blue box contains instructions to drop a vector polygon file. Supported file formats listed are Shapefile (zip including .shp, .dbf, .prj, and .shx files) and GeoJSON (.json or .geojson).
- Start Date:** A date input field set to '01-01-2000' with a calendar icon.
- End Date:** A date input field set to '01-01-2023' with a calendar icon.
- Is Date Recurring?:** A checkbox that is currently unchecked.
- Select the layers to include in the sample:** A list of layers with a search bar. The selected layer is 'Terra MODIS Net Evapotranspiration Gap-Filled (ET & LE) MOD16A2GF.061, 500m, 8 day, 2000-01-01 to Present'.
- Selected file (baeia_estacao84):** A map view showing a green polygon over a satellite image of a coastal area. The map includes a scale bar, a compass, and coordinates (Lat: -25.602, Lon: -48.645).
- Selected layers:** A list of layers with their respective resolutions and time intervals. The selected layers are 'ET_500m' (500m, 8 day) and 'PET_500m' (500m, 8 day).

AppEARS - data from Estonia, Afghanistan and Brazil



The screenshot shows the AppEARS web interface. At the top, there's a header with the NASA EarthData logo, 'Other DAACs', and navigation links for 'Extract', 'Explore', and 'Help'. A user profile 'emiliomercuri' is logged in. A blue banner at the top of the main content area contains a warning: 'Some MODIS/Terra Snow Cover v6.1 (MOD10A2) tiles are currently unavailable to AppEARS. Requests containing MOD10A2 data may result in processing errors.' Below this, the 'Explore Requests' section is active. It shows 'Showing requests 1 - 17 of 17' with pagination controls. A table lists the requests with columns for Request, Type, Status, Details, Date Submitted, and Date Completed. The first two requests are 'In Progress', while the last three are 'Expired'. Each row has icons for viewing details, downloading, and deleting.

Request	Type	Status	Details	Date Submitted	Date Completed
Nhundiaguara	Area Sample	In Progress		03-10-2023 3:54:45 PM GMT-3	
Ahja watershed MOD16 ET & PET	Area Sample	In Progress		03-10-2023 3:50:05 PM GMT-3	
Temperatura_Kalli	Area Sample	Expired		01-19-2023 11:31:58 AM GMT-3	01-19-2023 12:48:25 PM GMT-3
Reola watershed MOD16 ET & PET	Area Sample	Expired		01-11-2023 11:32:41 AM GMT-3	01-11-2023 5:13:17 PM GMT-3
Kalli watershed MOD16 ET & PET	Area Sample	Expired		01-11-2023 9:36:47 AM GMT-3	01-11-2023 10:20:39 AM GMT-3

Application for Extracting and Exploring Analysis Ready Samples (AppEARS) =

<https://appears.earthdatacloud.nasa.gov/>

Download the data

 AppEEARS Extract ▾ Explore Help ▾   emiliomercuri ▾

Download Area Sample

Request: Nhundiaquara ▸

Supporting Files

	Nhundiaquara-MOD16A2GF-006-metadata.xml	ISO 19115 Metadata	21.64 KB
	Nhundiaquara-granule-list.txt	URLs for all source data used in the extraction	111.68 KB
	README.md	Instructions and details about the request	24.47 KB
	Nhundiaquara-request.json	JSON file which can be used to create a new request	128.16 KB
	MOD16A2GF-006-ET-QC-500m-Statistics-QA.csv	Statistics for quality layers	90.84 KB
	MOD16A2GF-006-ET-QC-500m-lookup.csv	Lookup values for the quality bits	2.29 KB
	MOD16A2GF-006-Statistics.csv	Statistics for layers	309.03 KB



0 Selected Download ▾

☐	Name ↑↓	Size ↑↓
☐	 MOD16A2GF:006_500m_aid0001.nc	4.84 MB

1 - 1 displayed, 1 in total

Download the data - File formats, projections

- QGIS or Colab - Shapefile preparation -> zipfile (Aula 2)
- Data formats:
 - NetCDF (Network Common Data Form) version 4
 - GeoTIFF
 - CSV file (MOD16A2GF-006-Statistics.csv) - We will only use this one!
 - ET and PET comes in $\text{kg/m}^2/8\text{-day} = \text{mm}/8\text{-day}$
- Projection:
 - Geographic
 - Datum: WGS84

Thanks! Let's code!

Google Colab

What we will do:

- Import the data into COLAB
- Process it to daily data
- Compare ET from Afghanistan Brazil and Estonia

